Is Your Database Meeting Your Needs?

Databases are based either on the ACID or the BASE model. The former ensures consistency of transactions while the latter guarantees availability. Both models are governed by the CAP theorem. These key concepts shape your database decisions. Let's explore them.

Data is the new oil and often referred to as the heart of any organisation. After all, every organisation relies on data to make both short-term and long-term decisions. Choosing the right database is crucial and challenging, as your data resides in it. There are various types of databases, including SQL and NoSQL, centralised and distributed, in-memory and persistence storage — each with advantages and disadvantages.

In this article, we will explore three important concepts: the ACID database model, BASE database model and the CAP theorem. These concepts define different dimensions such as consistency, guaranteed transactions, stable distributed system, and more. There is no single best concept, as each carries its own strengths and challenges. Let's delve deeper into how each concept ensures data consistency, availability and transaction reliability. Also, let's explore how each concept is different from another.

The ACID database model

For any database, whether it is SQL, NoSQL or GraphDB, ACID principles are fundamental when it comes to handling transactions.

Before we proceed, let's first understand what a transaction is — it is a collection of queries that are treated as a single task. So, we need to run all or none at all. If something inbetween fails either intentionally or unintentionally, all changes made within the transaction need to be rolled back to the state just before the transaction began.

For example, XYZ transfers ₹100 to ABC. The transaction is as follows:

- Transaction begin
- Read balance from XYZ
- Proceed only if XYZ > ₹100
- XYZ debit ₹100